

Title:
Governing Unstructured Data

Subtitle:
Integrating Multi-Protocol Storage, Cross-Platform Access Control, and a NoSQL Database for Data Governance and Storage Compliance

Byline:
By Steve Hoenisch,
Likewise Software

max is 4000 WORDS: FOCUS EVERY ASPECT OF THIS ON TECHNICAL ANGLE in solving people-process problem and keep it short.

#EXECUTIVE SUMMARY

This white paper proposes a storage architecture to solve a lingering problem in data governance: controlling sensitive unstructured data. The paper argues that a multi-protocol file server or NAS system integrated with cross-platform access control establishes a secure framework to consolidate, protect, and audit unstructured data, reducing the need to develop complex interdepartmental processes that can take years to put in place.

To measure the effectiveness of data governance processes and security policies, the architecture includes a high-performance activity monitoring system to track unstructured content in a security-aware context of user identities, access patterns, and file events. An integrated application for managing sensitive folders and files empowers users to govern unstructured data.

In addition, the storage technology integrates big data analytics with file server technology to heighten the strategic importance of IT: Tasks such as discovery, classification, and data governance become easier. Such services can increase competitive advantage and revenue while reducing costs and mitigating risks.

But the real value of the architecture is that it enables you to get around several long-standing barriers to enterprise data governance: autonomous departments, lack of consensus, and complex processes. Using technology to bypass these barriers can ameliorate the need to change your organization's behavior and processes while enabling the data governance program to fulfill its promise of managing risk, guaranteeing data integrity, and creating value.

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INTRODUCTION

Data governance¹ refers to the overall management of the availability, usability, integrity, and security of the data employed in an enterprise. A sound data governance program includes a governing body or council, a defined set of procedures, and a plan to execute those procedures. In practical terms, that means putting personnel, policies, procedures, and organizational structures in place to make data accurate, consistent, secure, and available to accomplish the business objectives. It can take on special importance because of the legal requirements and compliance regulations that it must fulfill in the face of rapidly expanding volumes of unstructured data.

Unstructured data, industry analysts say, is growing faster than all other types of data and will increase by as much as 800 percent during the next five years. Respondents to an industry survey by the Aberdeen Group estimated that 40 percent of their sensitive data is in

unstructured formats such as PDF and Microsoft Word and Excel.

"Rising volumes of content and an expanding array of electronically stored information combined with increasing litigation, staff shortages, limited integration with key applications, and other factors lead to low confidence and considerable expense during the eDiscovery process. Only 28% of message archiving stakeholders report that they are very confident that if challenged their organization could demonstrate that their electronically stored information (ESI) is accurate, accessible, and trustworthy" March 17, 2010

Regulatory And eDiscovery Demands Drive

A Growing Message Archiving Market

Q1 2010 Global Message Archiving Online Survey

by Brian W. Hill At Forrester.

THESE NOTES ARE KEYS.

Any data governance program, including system and processes, is specific to your organizational needs. This white paper lays out an abstract architecture for data governance technology that can help meet the data governance needs of organizations in the context of protecting, auditing, controlling, and compliance around unstructured data. Because the architecture is abstract, it allows the technology to address at least three focus areas within the field of data gov, as set forth by the Data Governance Institute
http://datagovernance.com/fc_focus_areas_for_data_governance.html:

- *Policy, Standards, Strategy

- *Privacy, Compliance, Security

- *Architecture Integration, Analysis

- *Data Warehouse and Business Intelligence

Again, the focus here is on data governance for sensitive unstructured data. Don't worry about non-sensitive data when you build your data governance program: A common inhibitor, says Forrester analyst Rob Karel in Data Governance: What Works And What Doesn't, is unrealistic scoping. "Don't assume that all data must be governed equally. It needn't. Only the data supporting your most critical business processes and decisions require the highest level of governance."

"Organizations typically find it easier to focus on compliance first, and then gradually expand efforts to include risk management and governance. It is important to note, however, that governance activities will happen, whether planned or not, and that lack of planned governance and rigorous risk management can have serious consequences for the business."

MS DATA GOV PART 1

Not just compliance and security, either, but freedom of users to rapidly access intellectual property that helps foster collaboration and innovation:

"Most respondents report they either do not have or are not aware of governance procedures in place for unstructured data, suggesting that much of this kind of data is haphazardly being distributed and managed across enterprises."--By Joseph McKendrick, Research Analyst, Produced by Unisphere Research, A Division of Information Today, Inc, June 2011

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#OUTCOMES: WHY GOVERN UNSTRUCTURED DATA?

Organizations undertake initiatives to govern unstructured data for reasons that vary by organization, company or institution, industry, and objective. Some common reasons or

targeted outcomes are as follows, many of which tend to be interrelated:

<!-- bulleted list in this section with first word and trailing period set in bold for emph -->

*Compliance. Develop a data control framework to help demonstrate compliance with regulations, legal requirements, industry standards, and internal security policies.

*Security and confidentiality. Protect stored secrets, intellectual property, private and confidential data, and other information assets.

*Risk mitigation. Mitigate risk of security breeches, data loss or exposure, fraud, noncompliance, and costly legal problems.

*Access and availability. Provide rapid access to intellectual property and business-enabling data to drive business objectives and to foster innovation, revenue, and the creation of value.

*Quality and integrity. Ensure that important data is trustworthy, accurate, and up to date to support transactions, decision making, research, and other business functions.

*Cost reduction. All the elements in this list play a role in reducing costs in a variety of ways, from avoiding duplication of data or documents to keeping down the costs of storing and managing data. Decisions based on accurate information can radically reduce cost.

*Innovation, value creation, profit. All the outcomes in this list, of course, point to the bottomline of businesses: increasing revenue.

<!-- SECTION HEAD -->

#INHIBITORS: BARRIERS TO GOVERNING UNSTRUCTURED DATA

Although the outcomes that a data governance program is intended to bring about are, at least at a high level, straightforward enough, achieving those outcomes in large organizations and complex enterprises is another thing altogether. Here are some common inhibitors to developing a data governance program for unstructured data. Most of them fall into one or more of three groups: cross-enterprise complexity, complex organizations, and complex data. Most of the inhibitors translate into a requirement -- requirements that I'll return to later to identify the extent to which they can be solved by the right technical architecture for unstructured data. SEE UW PIECE!

<!-- SECTION SUBHEAD -->

COMPLEX ORGANIZATIONS

*Lack of awareness and sponsorship. Administrators and users are frequently unaware of regulatory requirements for sensitive data as well as the importance of data quality to decision makers. Without automated systems to force adherence and to monitor for lapses, users may inadvertently subvert data governance requirements. Requirement: Advocate for the importance data integrity and security; empower employees to take ownership of data governance tasks with automated tools.

*Lack of ownership. A lack of awareness and sponsorship can result either in a lack of data stewards and custodians or in stewards who are not empowered to adequately govern data. "Best-in-class organizations are 2.2-times more likely than laggards to assign responsibility to *someone*," the Aberdeen Group says in an industry survey on securing unstructured data. The fallout from the mantra that data quality is everyone's responsibility is that the task goes unassigned. Requirement: Assign the task of governing important unstructured data to one or more managers and give them technology

that makes it easy for them to govern data.

***Lack of consensus.** Interdepartmental disagreements about how to process and store common data can undermine an enterprise-wide data governance program.

Requirement: Explore the prospect of using technology to skirt the need for interdepartmental consensus.

***Independent departments.** Departments empowered to act autonomously for business reasons can result in inconsistencies in shared data as well as in data silos segmented by protocol: The engineering department stores unstructured data on NFS-based Unix file servers, while the research department uses CIFS-based Windows file servers. Data silos inherently hinder data governance because it is hard to apply universal controls across protocol boundaries. **Requirement:** Deploy multi-protocol file servers or NAS systems to eliminate data silos and foster universal controls.

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COMPLEX DATA

***Application data is increasingly going to file servers.** Application workloads are migrating to filers because it is easier to provision file-based storage than block-based storage. The application data further heightens the need for exception-based methods of monitoring and auditing. **Requirement:** Exception-based management of unstructured data and application data stored on file servers.

***Incompatible or conflicting security descriptors.** The security policies that are defined in security descriptors for files and directories are inconsistent across Windows and Unix systems. A Windows computer uses ACL; a Unix or Linux computer uses POSIX. Inconsistent security descriptors can limit the ability to access stored files across different storage platforms. It can also result in file security being changed or removed, which can undermine a data governance program's objective of mitigating risk and complying with regulations.

***Unrealistic scoping and different notions of what data is important.** When the scope of your data governance program is too wide, the program can bog down in managing data that is not sensitive or is not critical to your operations or objectives. Unrealistic scoping gets in the way of discovering and classifying important data. Meantime, different data owners might have different notions of what data is important, compounding the scoping problem. **Requirement:** A mechanism for marking data as important or sensitive and a method for tracking it in storage.

***Data without metadata.** Unstructured data either contains no metadata or only limited, application-specific metadata, a void that makes it harder to discover sensitive information and classify it for protection.

<!-- SECTION SUBHEAD -->

CROSS-ENTERPRISE COMPLEXITY AND COMPLEX TECHNOLOGY

***Silos of unstructured data.** Because of distributed environments, virtualization, disparate operating systems, and different protocols, unstructured data ends up in file servers segmented by protocol: Unix data goes to NFS-based storage devices, while Windows data goes to CIFS-based storage devices. For data governance, data silos are a major structural inhibitor with a pervasive, cascading effect: They hinder cost-saving storage consolidation, centralized monitoring, and universal governance. Data silos, in effect, can obstruct the development of a common framework to govern data. More importantly, data silos pose a barrier to the second major structural inhibitor: a common security model and uniform access control policies. **Requirement:** Consolidate data silos to a multi-protocol file server.

*Inconsistent security policies. A heterogenous network not only creates data silos, but also renders cross-platform access control difficult. In an enterprise, there are typically multiple identity management and access control systems, such as NIS or LDAP for Unix file servers and Microsoft Active Directory for Windows file servers. Multiple identity management and access control systems are a major barrier to ensuring quality, integrity, availability, confidentiality, and compliance while mitigating risk. Requirement: Implement a cross-platform identity management and access control system that can impose consistent security policies.

*Information lifecycle management. In a mixed network, the problems associated with disparate access control systems and different storage protocols combine to sabotage information life cycle management (ILM) and records management. Cross-platform complexity inhibits the tasks of identifying, collecting, classifying, and controlling sensitive information.

*Monitoring and auditing problems. Monitoring the effectiveness of your data governance program across silos is another obstacle that stems from data silos. Compliance auditing falls victim to the same limitations: Monitoring and auditing technologies such as sniffers, agents, and shims can degrade the performance of your network and storage system. They also don't scale to manage the flow of data and metadata that is required for generating near-real-time contextual awareness of unstructured data.

More importantly, however, monitoring technologies like SIEM tools are not closely tied to the identity management system -- blocking standardized audits that link users to identities and organizational roles.

*Lack of comprehensive reports with visibility across silos. Storage silos and multiple access protocols impede enterprise-wide reports that aggregate data for compliance. Because there are different identity management systems for different silos, the reports cannot associate user accounts with their identities in the organizations, leaving you unable to demonstrate compliance with some regulations.

*Heterogenous storage systems undercut best practices. There are commonly referenced best practices and guidelines to manage risks, which vary by industry but can include COBIT and ISO/IEC 38500.

The complexity, heterogeneity, and duplication of storage platforms and access control systems raises IT costs and impedes data governance. Implementing technology that can help bridge the cross-platform problems can give you the leverage to implement a successful data governance program and transcend the need to solve some of the problems associated with complex organizations and complex data.

The next section proposes an architecture that lays a technological foundation for helping put in place many of these enablers. The subsequent section evaluates how the technology solves the problems and supports the use cases laid out at the beginning of the paper.

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#ARCHITECTURE AND FEATURES

The following components provide the architecture for a file server that supports data governance of unstructured data across storage protocols.

<!-- bulleted list -->

- *A multiprotocol, cross-platform file server or NAS system that supports CIFS and NFS to

allow connections from both Windows and Unix computers.

- *An integrated authentication engine that can authenticate and authorize users with Active Directory, NIS, or LDAP.

- *An integrated application for marking and tracking sensitive folders and files.

- *A secure event monitoring subsystem with collectors and forwarders that record, manage, and transmit file activity events, including events from other file servers.

- *A NoSQL database for event aggregation, processing, and advanced analytics.

- *A SQL data store for reports.

- *An auditing and reporting console.

- *An events dashboard for near real-time monitoring.

<!-- pu diagram of our architecture -->

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<!-- NEW MAIN SECTION -->

#HOW THE ARCHITECTURE AND TECHNOLOGY ENABLE DATA GOVERNANCE

This section explains how the architecture set forth above solves the problems and fulfills the technical and feature requirements of a data governance program for unstructured data.

<!-- SECTION SUBHEAD:: -->

MULTIPROTOCOL FILE SERVER ACCESSIBLE BY WINDOWS AND UNIX

At the foundation is a file server that is multi-protocol and cross-platform: It natively supports both the SMB/CIFS and the NFS protocols, making it usable simultaneously by Windows and Unix or Linux clients. A cross-platform, multiprotocol file server solves the interoperability problem that often separates the data of Unix users from the data of Windows users, providing a consolidated approach to storage for users of all types of computers. It enables a data governance program to address the unstructured data that resides in silos differentiated by operating system.

Cross-platform incompatibilities have also been a hindrance to applying a uniform set of security policies and to monitoring data across disparate platforms. In the past, just as there have been different, incompatible storage systems for Unix and Windows users, there have also been different, incompatible identity management systems for Unix and Windows users. Unix clients have tended to use NIS or LDAP, while the de facto standard for Windows clients is Microsoft Active Directory.

In this architectural schematic, therefore, the file server includes an integrated identity management service to authenticate users with Active Directory, NIS, or LDAP -- a component that, when combined with the multiprotocol file server, lays the architectural foundation for solving many of the problems in governing unstructured data.

The overall result is twofold. First, it frees your data governance program from the bounds of platform-specific storage, enabling you to govern all the stored data from a single system. Second, it secures the unstructured data by applying a common security model to it, enabling you to associate data access with user identities and roles for security monitoring and compliance.

<!-- SECTION SUBHEAD:: -->

SECURE CROSS-PLATFORM ACCESS CONTROL FOR UNSTRUCTURED DATA

The integrated identity service delivers direct, authoritative, robust security to control access by user or group, including blocking all external users and allowing only those internal users and groups that you specify. Simply put, you can control access to sensitive unstructured data and, as described below, use the built-in auditing framework to demonstrate those controls for compliance.

The file server's tight integration with the identity service also gives you visibility into the entitlements and permissions that are used to access and modify files. Because the identity service is integrated with the file server, you can both secure access by entitlement and generate reports to prove the entitlements secure the data.

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CLASSIFY AND TRACK SENSITIVE FILES TIED TO IDENTITIES AND OWNERS

The integrated identity service lets data stewards and custodians mark sensitive files, associate them with the identities of their owners, and track changes by user or group. Records managers who are charged with managing confidential information in unstructured files can limit access to specific users and groups and map changes to the files to those users and groups.

<!-- SECTION SUBHEAD:: -->

COLLECT ACCESS DATA AND FILE EVENTS FOR ANALYTICS AND REPORTS

The event collectors and forwarders form the event monitoring subsystem. On the file server, the event collectors record information about moving, copying, reading, modifying, or deleting directories or files. The collectors also capture changes to security descriptors.

Over a secure connection, the event forwarders send the file events on to the NoSQL database, where they are stored in a highly flexible format that later allows them to be manipulated for a variety of purposes, including big-data analytics, business intelligence, and forecasting.

The NoSQL database is the basis for a powerful, flexible analytics engine that can aggregate and correlate content types, sensitivity levels, modification attempts, security descriptors, user entitlements, and access patterns. An analytics system can, in turn, use data about past access patterns and file activities to hypothesize about future patterns of data storage. The inferences of an analytics system can help identify files that might contain sensitive or problematic data and need to be flagged for inspection or tracking. The NoSQL system, meanwhile, interfaces with a SQL Server database that segments frequently used data into columns and rows for reports, including custom queries.

The SQL Server also makes the data available to the dashboard for near real-time display of file events, especially exceptions, so that they can be acted on to deal with threats, data breaches, compliance violations. For extra security, the solution can be easily combined with interoperable storage encryption.

The auditing and reporting console can be used to create custom reports or reports based on templates to fulfill compliance regulations such as SOX, HIPAA, HITECH, and PCI DSS. You can choose your own data points. A simple interface makes it easy to create new reports, including custom reports. The reports let you audit file access and events by directory or server.

For internal and external threat monitoring, the dashboard displays file events correlated with permissions in near real-time. With the dashboard, you can proactively monitor user access and changes to sensitive files or any other files and directories that you want to track. It enables you to respond to data faults, policy violations, potential breaches, or other security incidents.

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PERFORMANCE

Millions of file events can easily overwhelm the network and the monitoring system.

Because of the number of events that are generated in an enterprise as a multitude of users access and modify files, performance is a requirement that must be considered up front -- but all too frequently is not, and it is only after implementation that performance issues emerge: networks slow down, databases overwhelm disk space, dashboards freeze.

The performance of the event monitoring system plays a key role in how fast and efficiently many of the end-user components that rely on the events will function. To be expedient and relevant, exception monitoring depends on how fast events are collected and correlated. The auditing tools also rely on the performance of the system to quickly produce up-to-date reports.

To ensure that events do not consume too much network traffic or bog down systems, monitoring ultimately should take place as part of the file server. When the monitoring is handled by the file server and is built with performance in mind, it can ensure that the system scales efficiently to deliver high performance in high-traffic environments.

To evaluate performance, you must take various aspects of the architecture and its interconnections into account. Start with the database: Is it highly scalable? What about the connections to it? Will the forwarding of events degrade network traffic? If the network connection goes down, does the system include scalable solutions to continue collecting and queuing data for forwarding later, without data loss?

Then there's the question of the performance of the entire stack. Consider the following potential performance bottlenecks based on the architecture described above:

<!-- DID YOU TIE UP LOOSE ENDS OF THIS LIST; IS EACH OF THESE ADDRESSED BELOW POSITIVELY?? also: can you check that you tied up loose ends in actually solving the uses cases and probs you laid out earlier?? are they sufficiently covered?? -->

- *Collectors, especially when there are potentially thousands of events taking place simultaneously on a large file server.
- *Forwarders, especially when the network connection goes down and events must be forwarded later.
- *Submitting queries for reports to the database.
- *Continually pulling updates from the database to populate the dashboard with events.

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FOLD THIS INTO SECTION ABOVE -- IT THE SAME THING
DISCUSSION: HOW ARCHITECTURE SOLVES MANY OF THE PROBLEMS
laid out earlier and takes some onus off people, processes, policies, interorg communication,
and departments; REFERENCE BOTH MODELS, PROBLEMS, AND USE CASES

SIMPLIFIES MODEL
SOLVES PROBLEMS
ENABLES OUR USE CASES

PLS REF EACH PROB SPECIFICALLY THIS TIME; revisit each use case to show how solved

KEEP IT SHORT!

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#CONCLUSION

This white paper prescribes a storage architecture to solve a lingering problem in data governance: controlling sensitive unstructured data. The paper argues that a multi-protocol file server or NAS system governed by cross-platform access control establishes a secure framework to consolidate, protect, and monitor unstructured data. The storage architecture helps transcend several long-standing barriers to enterprise data governance: autonomous departments, lack of consensus, and complex processes.

To measure the effectiveness of data governance processes and security policies, the architecture includes a high-performance file activity monitoring system to track unstructured content in a security-aware context of user identities, patterns of access, and file change events. An integrated application for marking sensitive folders and files empowers users to govern unstructured data.

In addition, the storage technology integrates big data analytics with file server technology to provide services like security monitoring, classification, and data governance. Such services can increase competitive advantage and revenue while reducing costs and mitigating risks.

The result is that the storage architecture can overcome more problems in data governance while reducing costs, requiring less effort, and ameliorating the need to change your organization's behavior.

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KEYS AND NOTES AND CITATIONS ETC

LEAPING INTO THE FUTURE WITH DATA GOVERNANCE

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"This approach -- the most advanced of the four, and also the most complex and demanding - involves comprehensive integration of not only database security, but also data security, within the domain of enterprise information management or governance. Security and risk management are addressed as a component of an enterprise business initiative in a cyclical approach that addresses discovery and valuation; end-to-end documentation and business context awareness; automated end-to-end awareness and control; and continuous improvement, metrics, reporting, real-time enforcement and control. Gartner is seeing more adoption of this approach, which confirms our long-standing view that information security and risk management need to be better integrated into the business. This is not simply a matter of selecting or implementing a security product, but rather of building business processes around business needs and making the management of security and risk related to those processes and needs a business requirement."

"This robust approach is beyond the capabilities of most enterprises today, and only one vendor, IBM (through the technology acquired via its 2009 acquisition of Guardium), offers technology capable of supporting this approach by leveraging the other tools in its information management practice. (Other vendors are known to be investigating this approach.) It is important to recognize that the data/information governance approach extends well beyond

database/data security and, in fact, well beyond security and risk management. It enables improved data and process availability, makes it possible to locate and aggregate data for analytics and reporting purposes, and allows real-time enforcement and control, as well as continuous process improvement. For all these reasons, this approach holds the promise of significant business benefits and potential cost savings for those few enterprises that have reached the necessary stage of maturity. Enterprise Data Security The next most-mature approach takes an enterprisewide view of all data -- structured and unstructured -- that exists in the core of the typical enterprise and addresses the protection of that data throughout its life, including identification, risk assessment, access controls and controls enforcement across all data storage platforms. This approach is best characterized by offerings from Imperva and Netezza. Imperva treats DAM as a component of a data protection and risk management function. Netezza's Mantra offering provides support for structured and unstructured"

Enterprise Data Security The next most-mature approach takes an enterprisewide view of all data -- structured and unstructured -- that exists in the core of the typical enterprise and addresses the protection of that data throughout its life, including identification, risk assessment, access controls and controls enforcement across all data storage platforms.

This approach is best characterized by offerings from Imperva and Netezza. Imperva treats DAM as a component of a data protection and risk management function. Netezza's Mantra offering provides support for structured and unstructured data and bundles DAM with an analytics engine and data-warehousing component as part of an enterprise data security program. Under this scenario, structured and unstructured data are located, categorized and classified, and access is monitored, not only in databases, but in the data center platform in the data center. This requires working closely with the business, performing classification and evaluation -- some of this can be done automatically -- and then building a complete data protection framework. One or more vendors may be adding capabilities for activity monitoring of unstructured data, to enable enterprises to understand what is happening with their Windows or Unix file shares, for example. Gartner believes this is an important potential development, and one that enterprises considering DAM solutions should follow closely, because a narrow focus on structured data is a long-standing weakness of DAM technology. Enterprises should also consider alternatives, such as security information and event management (SIEM) or some combination of SIEM, data loss prevention (DLP) and DAM offerings to support data security initiatives.

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FROM LW-MS PRESS RELEASE::

In computer networking, SMB operates as an application-layer network protocol mainly used to provide shared access to files and miscellaneous communications between nodes on a network. It also provides an authenticated inter-process communication mechanism.

QUOTE FROM MS:: "Safely, reliably and efficiently managing storage, while providing high performance and scale, is becoming increasingly important. SMB 2.2 will provide the protocol foundation for delivering these capabilities in Windows Server 8."

"For many IT organizations, block storage has been the only option for provisioning

Microsoft application workloads. Block storage is complex and costly to administer," said Ken Cheney, vice president of marketing and business development, Likewise. "With Windows Server 8, workloads running on Hyper-V, as well Microsoft applications such as IIS and SQL Server, will be able to take advantage of the SMB 2.2 protocol, which aims to provide significant performance, reliability and scalability improvements unattainable with prior versions of SMB. The new SMB 2.2 has the potential to reduce the complexity and costs associated with block storage by enabling both application and user workloads to access file-based storage using NAS devices."

The Likewise platform, Likewise Storage Services, is available for licensing under commercial terms by OEM vendors interested in NAS-enabling their Linux- or Unix-based storage devices. It offers a consistent security model for file-based access and cross-platform, integrated storage across physical, virtual and cloud environments. The software provides integrated identity and access management, as well as both access to and control of a company's intellectual assets, from Windows, Unix and Linux systems.

Protocols supported by Likewise Storage Services include SMB/CIFS 1.0, 2.0, 2.1 and NFS 3.0. Consistent security policy, data management, audit and reporting are available across these protocols for security and data governance. Pending protocol support for Likewise Storage Services includes SMB 2.2, NFS 4.x / pNFS, HTTP/s, WebDAV, and a REST API. Likewise Storage Services supports multiple directory service providers including Active Directory, LDAP and NIS. The modular, API-driven architecture of Likewise Storage Services delivers reliable and scalable performance for a variety of workloads and allows for easy integration with most proprietary vendor file systems.

About Likewise

Likewise makes an integrated software platform for identity, security and storage used by market-leading storage vendors including EMC and HP. Likewise enables organizations to provide both access to and control of their data across mixed network environments. More information is available at the company's website, www.likewise.com.

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A common inhibitor, says Forrester analyst Rob Karel in *Data Governance: What Works And What Doesn't*, is unrealistic scoping. "Don't assume that all data must be governed equally. It needn't. Only the data supporting your most critical business processes and decisions require the highest level of governance."

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EXAMPLES FROM MS GUIDE TO DATA GOV PART 1::

According to DataLossDB, a nonprofit organization that tracks data breach incidents, more than 218 million records containing individuals' personal data were compromised in 436 incidents in 2009.⁵

A study by the Ponemon Institute of organizations that suffered one or more data breach incidents in the United States in 2008⁶ found that the average cost incurred by the organizations was US\$6.6 million (not including long-term litigation costs and the resulting penalties). The study also found that these organizations lost an average 3.6% of its customer base as a result of the incident(s). For financial institutions, the average loss was 6%.

In January 2009, U.S.-based payment processor Heartland Payment Systems disclosed what is possibly the largest data breach on record. An identity theft ring stole an estimated

130 million credit and debit card records from the company's databases.⁵ The CEO estimated that the company spent close to US\$32 million in the first half of 2009 in IT and legal costs related to the breach.⁷ Heartland's market value dropped by 75% in the two months following the announcement of the incident.

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NOTES QUOTES AND EXAMPLES FROM MS GUIDE TO DATA GOV PART 1::

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2 American Institute of Certified Public Accountants (AICPA) and Canadian Institute of Chartered Accountants (CICA), "Generally Accepted Privacy Principles,"

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3 Lyman, P. et al., "How Much Information?" University of California, Berkeley, 2003.

www2.sims.berkeley.edu/research/projects/how-much-info-2003.

4 Note that the term "confidential data," used throughout this document, encompasses personal data (under the European Commission definition), personally identifiable information (PII), intellectual property (IP), trade secrets, and all other types of information for which an organization might want to maintain confidentiality, integrity, and availability.

5 DataLossDB, <http://datalossdb.org/>.

6 Ponemon Institute, "Cost of a Data Breach," Feb. 2009, <http://www.encryptionreports.com>.

7 "Heartland CEO on Data Breach: QSAs Let Us Down," CSO Magazine, October 2009.

8 Joshua Gomez, Travis Pinnick, and Ashkan Soltani, "KnowPrivacy," June 1, 2009, www.knowprivacy.org/report/KnowPrivacy_Final_Report.pdf.

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"Data stewards are responsible for ensuring effective control and use of data assets and exercising a series of functions assigned to them by the data governance organization. Additional details on the data steward's role and responsibilities are provided in the second paper in this series."

"data steward A business leader or subject matter expert who is accountable for 1) the identification of operational and business intelligence data requirements within an assigned subject area, 2) the quality of data names, business definitions, and domain values within an assigned subject area, 3) compliance with regulatory requirements and conformance to internal data policies and data standards, 4) application of appropriate security controls, 5) analyzing and improving data quality, and 6) identifying and resolving data related issues.²⁴"
24 The DAMA Dictionary of Data Management, 1st Edition, 2008.

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SIDEBAR: GAUGE YOUR LEVEL OF DATA GOVERNANCE MATURITY FOR UNSTRUCTURED DATA IN MIXED NETWORKS

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10 short questions pulled out of discussion above.

MODIFY AND AUGMENT MODEL TO ADDRESS UNSTRUCTURED DATA IN MIXED NETWORKS::

This section applies the IBM Data Governance Maturity Model to evaluate how the architecture and technology outlined above can help improve data governance for unstructured data.

gauging your level of maturity against a model

Models of data governance provide a method against which you can gauge your level of maturity in how you handle and control important business data.

There are a variety of abstract models of data governance. One of particular note is that of IBM, the IBM Data Governance Maturity Model, developed by its data gov council that includes. This looks at the IBM Data Governance Maturity Model.

1. INFOSEC: Technologies for Addressing Infosec Triangle

To gauge information security, you can evaluate the extent to which the following four

technology domains provide protection of unstructured data for the three aspects of the infosec triangle:

Secure infrastructure
Identity and access control
Data protection
Auditing and Reporting

2. IBM 11 DISCIPLINES OF DATA GOV MATURITY: THIS PART BECOMES SIDEBAR!

1. Organizational Awareness:
2. Stewardship:
3. Policy:
4. Value Creation:
5. Data Risk Management:
6. Security / Privacy / Compliance
7. Data Architecture
8. Data Quality
9. Business Glossary / Metadata
10. Information Lifecycle Management
11. Auditing and Reporting

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"All storage compliance involves hardware and software elements, but there is no single approach or architecture to rely on. For example, disk-to-disk, disk-to-disk-to-tape and disk-to-disk-to-optical storage platforms all have potential applications in compliance, but regulations rarely, if ever, define storage implementations. Organizations are left to interpret the rules, formulate the requirements and establish the technologies to meet legal obligations. THIS IS THE BIG FLAW IN STORAGE COMPLIANCE -- organizations jump to a "vendor's solution" that promises compliance, without fully understanding the laws and their

impact. For example, a disk storage platform may offer reliability but might not provide the security or immutability demanded for your industry. "Buying an array for long-term retention is nice," says Jim Damoulakis, chief technology officer (CTO) at GlassHouse Technologies Inc. in Framingham, Mass. "But it doesn't make you compliant." Policies and processes must be implemented to manage storage compliance."--from <http://searchstorage.techtarget.com/feature/Data-storage-compliances-impact-on-storage-product-choices> (emphasis is mine)

"Workload is added in the monitoring, reporting and auditing capabilities needed to support storage compliance. "The thing that adds workload is that you now have to prove that you're doing it," Damoulakis says. In some cases, the cost can be substantial. Back in 2004, General Electric Co. revealed it spent about \$30 million in compliance costs just to meet SOX regulations."